Seonghyeon Kang

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PERSONAL DATA

- Birth : 4th Jan, 2002, in Republic of Korea
- Nationality : South Korean
- Family Status : Single
- Language : First Language Korean, Fluent in English
- Graduate Student of Computational Energy Bio/Soft Materials Lab

EDUCATION

2019.02. ~ 2023.08. **B.Sc. in School of Undergraduate**
Deagu Gyeongbook Institution of Science and Technology (DGIST, South Korea)
Magna Cum Laude

2023.09. ~ **M.Sc. in Chemistry**
Pohang University of Science and Technology (POSTECH, South Korea)
(PI : Prof. Chang-Yun Son)

WORK EXPERIENCES

- **Research Intern** at *Department of Emerging Material Science, DGIST* 2020.07 ~ 2020.07
  - ✓ Post-synthetic method of Cu_2(m-DOBDC) metal organic framework (Advisor : Prof. Jinhee Park)

- **Research Intern** at *Department of Energy Science and Engineering, DGIST* 2021.01 ~ 2021.01
  - ✓ Introduction to computational chemistry (Advisor : Prof. Stefan Ringe)
  - ✓ Basic experiments of electrochemical CO2 reduction (Advisor : Prof. Dae-Hyun Nam)

- **Undergraduate Research** at *Department of Energy Science and Engineering, DGIST* 2021.02 ~ 2021.12
  - ✓ Effect of copper oxidation states on CO2 electroreduction
    (Advisor : Prof. Stefan Ringe, Prof. Dae-Hyun Nam)

- **Research Intern** at *Department of Emerging Material Science, DGIST* 2022.01 ~ 2022.02
  - ✓ Stability of copper oxide in various potential and pH environment (Advisor : Prof. Stefan Ringe)

- **Research Intern** at *Department of Chemical Engineering, POSTECH* 2022.07 ~ 2022.07
  - ✓ Introduction to computational approach to heterogeneous catalysis (Advisor : Prof. Jeong Woo Han)

- **Research Intern** at *Department of Chemistry, POSTECH* 2023.01 ~ 2023.02
  - ✓ Preparation of polymer structure file and forcefield for MD simulation (Advisor : Prof. Chang Yun Son)

QUALIFICATION

- TOEIC 765

- TOEFL ITP 540(Expired)

AWARDS AND HONORS

- **Third Prize of Fame Lab competition**, DGIST 2020.12
- **2020 Dean's List**, DGIST 2021.05
- **2021 Spring Semester Dean's list**, DGIST 2021.09

SCHOLARSHIP AND FELLOW

- **Government Scholarship**, DGIST
  2019.02 ~ 2023.02
- **Sosun Sharing Fund for Graduate Student in Science and Technology**, Community Chest of Korea, Daegu
  2023.09 ~ 2024.08

TUTORING AND TA ACTIVITY

- **Tutoring** Multivariable Calculus (2020 Fall Semester, DGIST)
- **Tutoring** Organic Chemistry I (2021 Spring Semester, DGIST)
- **Tutoring** Physical Chemistry II (2022 Spring Semester, DGIST)
- **Tutoring** Organic Chemistry II (2022 Fall Semester, DGIST)
- **Tutoring** Physical Chemistry I (2022 Fall Semester, DGIST)
- **Teaching Assistant** Physical Chemistry II (2023 Spring Semester, DGIST)
- **Teaching Assistant** Physical Chem. and Instrumental Analysis Lab (2023 Fall Semester, POSTECH)

TALKS

- **2024 Winter Molecular Dynamics Seminar**
  Computational Energy Bio/Soft Materials Lab
  2024.01 ~ 2024.03

PUBLICATION

- († = The authors contributed equally to this work; * = Corresponding author.)
- Tetteh, E. B.†, Gyan-Barimah, C.†, Lee, H. Y.†, Kang, T. H., Kang, S., Ringe, S.*, & Yu, J. S.* (2022). Strained Pt (221) Facet in a PtCo@ Pt-Rich Catalyst Boosts Oxygen Reduction and Hydrogen Evolution Activity. *ACS Applied Materials & Interfaces*.
- Hydrophilic Photocrosslinkers as a Universal Solution to Endow Water-Affinity to Polymer Photocatalyst for Enhanced Hydrogen Evolution Rate (Submitted)